ASSIGNMENT-5: TUPLES

SRABANI KHUNTIA

GROUP-E1, ROLL NO:7

1. Create a tuple with different data and create an empty tuple using built-in function.

In [1]:
```
t=('Srabani', 7, 1.1, True)
print('Tuple : ',t)
t_empty=tuple()
print('Empty tuple: ', t_empty)
```

```
Tuple :  ('Srabani', 7, 1.1, True)
Empty tuple:  ()
```

2. Print the 3rd element from beginning and 4th element from end of a tuple.

In [2]:
```
t=('Srabani', 7, 1.1, True, '24BCSB03', 6, 2.6)
print(f'3rd element from beginning: {t[2]}\n4th element from the end: {t[-4]}')
```

```
3rd element from beginning: 1.1
4th element from the end: True
```

3. Count the number of items present in a tuple and then add two new items into it (one item using concatenation operator and another item using built-in function).

In [2]:
```
t=('Srabani', 7, 1.1, True, '24BCSB03', 6, 2.6)
print('Number of items present in tuple: ', len(t))
t+=(19,)
print(t)
```

```
Number of items present in tuple:  7
('Srabani', 7, 1.1, True, '24BCSB03', 6, 2.6, 19)
```

4. Find the number of occurrences for any repeated item present in a tuple.

In [13]:
```
t=('Srabani', 7, 1.1, True, 1.1, 1.1, '24BCSB03', 6, 2.6)
n=t.count(1.1)
print(f'Number of occurences of {1.1} is {n}')
```

```
Number of occurences of 1.1 is 3
```

5. Display the elements present in a tuple which are at the even position.

In [3]:
```
t=eval(input('Enter tuple: '))
print('Elements at even position are:', end=' ')
for i in range(len(t)):
    if (i+1) % 2 == 0:
        print(t[i], end=' ')
```

```
Enter tuple: ('Srabani', 7, 1.1, True, 1.1, 1.1, '24BCSB03', 6, 2.6)
Elements at even position are: 7 True 1.1 6
```

6. Remove the elements of a tuple which are present in the odd index position.

In [4]:
```
t=eval(input('Enter tuple: '))
t_updated=()
for i in range(len(t)):
    if i % 2 == 0:
        t_updated+=(t[i],)
t=t_updated
print('Updated tuple: ', t)
```

```
Enter tuple: ('Srabani', 7, 1.1, True, 1.1, 1.1, '24BCSB03', 6, 2.6)
Updated tuple:  ('Srabani', 1.1, 1.1, '24BCSB03', 2.6)
```

7. Remove the odd number elements from a tuple.

In [5]:
```
t=eval(input('Enter tuple: '))
t_updated=()
for i in range(len(t)):
    if t[i]%2==0:
        t_updated+=(t[i],)
t=t_updated
print('Updated tuple: ', t)
```

```
Enter tuple: (85,8,1,89,4,891,9,7,23,45,698,62,8,8,6,4)
Updated tuple:  (8, 4, 698, 62, 8, 8, 6, 4)
```

8. Reverse the elements of a tuple by using and without using a built-in function.

In [6]:
```
t=eval(input('Enter tuple: '))
t_rev1=t[::-1]
t_rev2=tuple(reversed(t))
print('Reversed without using buit-in function: ', t_rev1)
print('Reversed with using buit-in function: ', t_rev2)
```

```
Enter tuple: ('Srabani', 7, 1.1, True, 1.1, 1.1, '24BCSB03', 6, 2.6)
Reversed without using buit-in function:  (2.6, 6, '24BCSB03', 1.1, 1.1, True, 1.1, 7, 'Srab
ani')
Reversed with using buit-in function:  (2.6, 6, '24BCSB03', 1.1, 1.1, True, 1.1, 7, 'Sraban
i')
```

9. Sort the elements present in a tuple in descending order.

In [7]:
```
t=eval(input('Enter tuple: '))
t_sorted=sorted(t, reverse=True)
t=tuple(t_sorted)
print(t)
```

```
Enter tuple: (54,48,487,16,78,14)
(487, 78, 54, 48, 16, 14)
```

10. Take a tuple of mixed data types and find the square of each element.

In [8]:
```
t=eval(input('Enter tuple: '))
for i in t:
    print(i*i, end=' ')
```

```
Enter tuple: (1.1,8,5.1,6,0.2)
1.2100000000000002 64 26.009999999999998 36 0.04000000000000001
```

11. Given few numbers in a tuple, return 'True' if the first and last numbers are same.

In [9]:
```
t=eval(input('Enter tuple: '))
if t[0]==t[-1]:
    print('True')
else:
    print('False')
```

```
Enter tuple: (54,48,487,16,78,54)
True
```

12. Print only those elements present in a tuple which are divisible by 5.

In [10]:
```
t=eval(input('Enter tuple: '))
for i in t:
    if i%5==0:
        print(i, end=' ')
```

```
Enter tuple: (45,78,6,10)
45 10
```

13. Given a tuple of numbers, create another tuple such that it contains only the even numbers of the first tuple.

In [11]:
```
t=eval(input('Enter tuple: '))
t_even=()
for i in t:
    if i%2==0:
        t_even+=(i,)
print(t_even)
```

```
Enter tuple: (54,48,487,16,78,54)
(54, 48, 16, 78, 54)
```

14. Display the largest number of a tuple without using the built-in function max(). The user need to input the values in tuple from the keyboard.

In [12]:
```
t=eval(input('Enter tuple: '))
lar = t[0]
for i in t:
    if i > lar:
        lar=i
print(f"The largest number is: {lar}")
```

```
Enter tuple: (54,48,487,16,78,54)
The largest number is: 487
```

15. Replace the last value of the tuple with 100.

In [15]:
```
l=[(1,2,3),(4,5,6),(7,8,9)]
for i in range(len(l)):
    t_l=list(l[i])
    t_l[-1]=100
    l[i]=tuple(t_l)
print(l)
```

```
[(1, 2, 100), (4, 5, 100), (7, 8, 100)]
```